

## TAMIL NADU STATE BOARD - STANDARD 12 PHYSICS VOLUME 1

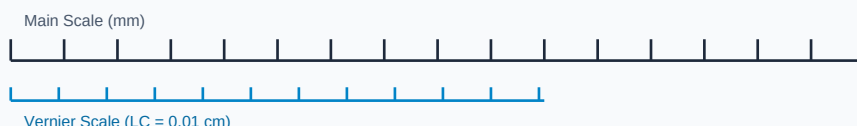
# CHAPTER 3: MAGNETISM AND MAGNETIC EFFECTS

## PREMIUM ACADEMIC STUDY SUITE &amp; EXAM GUIDE

|               |                                             |
|---------------|---------------------------------------------|
| Board:        | Tamil Nadu State Board (TNSB)               |
| Syllabus:     | Samacheer Kalvi Textbook Curriculum aligned |
| Standard:     | Class 12 English Medium                     |
| Subject:      | Physics Volume 1                            |
| Chapter Name: | Magnetism and Magnetic Effects              |

**Learning Objectives:**

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

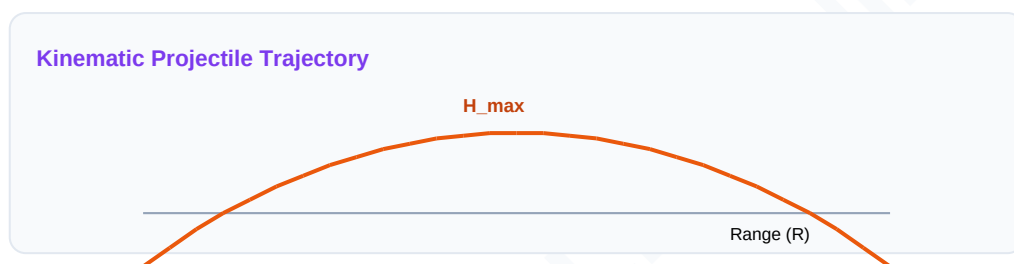
**Measurement Linear Scale Division**

## Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

| Term                          | Official Definition & Context                                                                      |
|-------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Magnetic Dipole Moment</b> | The product of pole strength and magnetic length of a magnet, representing torque reactions.       |
| <b>Magnetic Hysteresis</b>    | The lagging of magnetic induction behind the magnetizing field in a ferromagnetic material.        |
| <b>Gyromagnetic Ratio</b>     | The ratio of the magnetic dipole moment to the angular momentum of an electron orbiting a nucleus. |

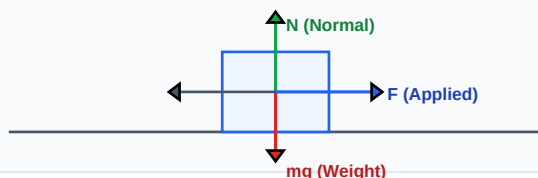
**Syllabus Exam Tip:** Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



## Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



### Essential Formulas, Laws & Identities:

- Biot-Savart:  $dB = (\mu_0 / 4\pi) * I * dl \times \hat{r} / r^2$
- Ampere's Law: Closed integral  $B \cdot dl = \mu_0 * I_{\text{enclosed}}$
- Lorentz Force:  $F = q * (v \times B)$  | Torque (loop):  $\tau = N * I * A * B * \sin(\theta)$
- Magnetic Field (solenoid):  $B = \mu_0 * n * I$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

## Worked Examples

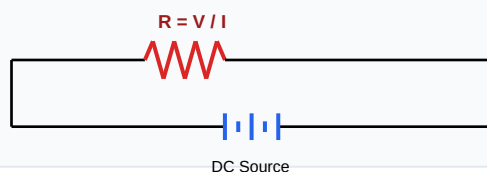
Review these standard board examination-style worked examples to learn proper step layout structures:

**Example 1: Calculate magnetic field at center of circular loop of radius 10cm carrying 5A current.**  
 ( $\mu_0 = 4\pi \times 10^{-7}$ )

**Step-by-step Solution:**

$$B = \mu_0 \cdot I / (2 \cdot R) = 4\pi \times 10^{-7} \cdot 5 / (2 \cdot 0.1) = 2\pi \times 10^{-7} \cdot 50 = 3.14 \cdot 10^{-5} \text{ Tesla.}$$

Schematic Electric Circuit Diagram



## HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

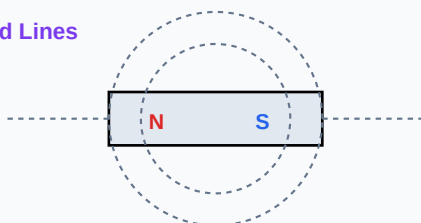
**HOTS Challenge 1: Show how a galvanometer can be converted into: (1) an Ammeter, and (2) a Voltmeter.**

**Analytical Solution Strategy:**

(1) Ammeter: Connect a very low shunt resistor  $S$  in parallel with galvanometer.  $S = I_g \cdot G / (I - I_g)$ .

(2) Voltmeter: Connect a high resistor  $R$  in series with galvanometer.  $R = V/I_g - G$ .

Magnetic Dipole Bar Field Lines



## Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

**Q1 (1-Mark):** State Biot-Savart Law in vector form.

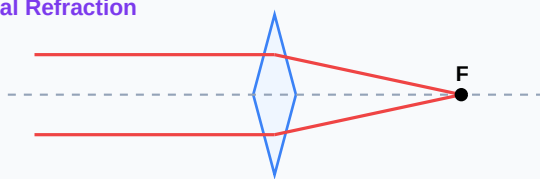
**Q2 (2-Mark):** Compare dia, para, and ferro-magnetic materials.

**Q3 (2-Mark):** State Ampere's circuital law.

**Q4 (5-Mark):** What is Cyclotron? Explain resonance condition.

**Q5 (5-Mark):** Calculate force between two parallel current-carrying conductors.

Convex Lens Optical Refraction



## Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

### Application Case: Diagnostic Systems

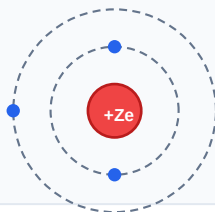
MRI scanners use superconducting solenoids to align hydrogen nuclei in body tissue scans.

### Application Case: Electric Motors

Industrial motor rotors spin using magnetic torque forces acting on coil loops.

**Cross-Curricular Link:** Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



## Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

### Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla,  $\text{cm}^3$ , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

### Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

**CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!**

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.  
Good luck!

#### Sinusoidal AC Voltage Waveform

